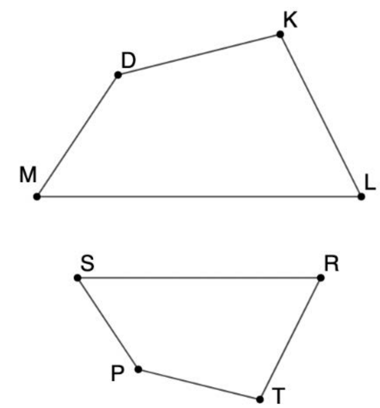
Geometry Name Answer Key__________
Unit 6 Date______________________
Lesson 11 Homework Period____________________

$DKLM$ and $PTRS$ are shown, where $DKLM \sim PTRS$ with $MD = 20$, $DK = 6z + 6$, $KL = 8w + 1$, $LM = 40$, $PT = 12$, $TR = 4w - 2$, and $RS = 16$.



1. Determine the value of each variable.

   $w =$ 3

   $z =$ 4

2. Determine the value of each side.

   $DK = 30$

   $SP = 8$

   $TR = 10$

3. What is the perimeter of $DKLM$?

   Perimeter $= 30 + 25 + 40 + 20 = 115$

Consider additional measurements of angles for $DKLM$ and $PTRS$ where $m\angle DKL = 150°$, $m\angle MDK = (15x - 25)°$, $m\angle SPT = (12x + 5)°$, $m\angle DML = (5y - 5)°$, $m\angle PSR = (4y + 4)°$.

4. Determine the value of each variable.

   $x =$ 10

   $y = 9$

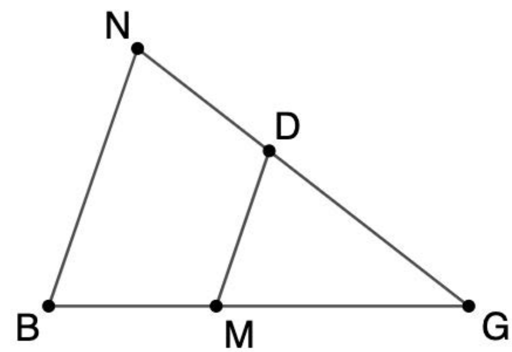
5. Determine the value of each angle.

   $m\angle MDK = 125°$

   $m\angle PTR = 150°$

   $m\angle PSR = 40°$

$\triangle NBG$ and $\triangle DMG$ are shown, where $\triangle NBG \sim \triangle DMG$ with $DM = 8$, $NB = 32$, $DG = 7$, $MG = 2y - 8$, and $BG = 4y + 4$.



6. What is the scale factor from $\triangle DMG$ to $\triangle NBG$?

$\frac{32}{8} = 4$

7. What is the value of the variable, y?

$\frac{32}{8} = \frac{4y+4}{2y-8}$

$32(2y - 8) = 8(4y + 4)$

$y = 9$

8. What is the length of BG?

$4(9) + 4 = 40 = BG$

Consider additional measurements of angles for $\triangle NBG$ and $\triangle DMG$ where $m\angle GNB = (8z + 3)°$, $m\angle DMG = (7w - 9)°$, $m\angle NBG = (6w + 3)°$, $m\angle GDM = (10z - 15)°$.

9. Determine the value of each variable.

$w = 12$

$z = 9$

10. Determine the value of each angle.

$m\angle GNB = 75°$

$m\angle GBN = 75°$

$m\angle DGM = 30°$